

Fahim Shahryer

AI/ML Engineer — MLOps
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Professional Summary

MLOps / AI Engineer with 2.5+ years building production-grade LLM systems, agentic AI architectures, and RAG pipelines serving 1000+ users. Specialized in multi-tenant SaaS, multi-LLM orchestration, and end-to-end ML lifecycle on AWS with FastAPI, LangGraph, and vector databases.

Professional Experience

MLOps Engineer

Sept 2025 - Present

Cloudcockpit

- Architected agentic backend of **Actionboard**, a multi-tenant project management platform where AI agents autonomously generate goals, action lists, and project plans from natural conversation – serving 5+ enterprise tenants with isolated knowledge bases
- Built **RAG pipeline** with hybrid global/tenant retrieval – automated PDF/DOCX ingestion via AWS S3, chunking, embedding, and semantic search on **PostgreSQL pgvector**
- Orchestrated 4 **LLM providers** with a custom multi-step reasoning framework (RAOARA) for cost-aware routing and structured agent decomposition
- Designed an extensible **MCP ecosystem** – integrated 7 internal MCP servers (exposing 20+ tools) and enabled tenants to connect their own custom MCP servers, dynamically expanding agent tool capabilities by 3x at runtime
- Deployed scalable infrastructure with **FastAPI, LangGraph, Neo4j, Redis**, and AWS (ECS, RDS, Cognito, ElastiCache) – with real-time SSE streaming and per-tenant token usage analytics

AI/ML Engineer

Jan 2025 - Sept 2025

Periscope Labs, Dhaka

- Built an **Agentic AI** framework for document validation using **RAG** and multi-LLM routing (OpenAI, Mistral) to auto-validate invoices and policy documents – reduced manual review effort by 70%
- Engineered an **admin dashboard** with an **annotation** pipeline that programmatically marks policy violations directly in source documents, improving auditor turnaround time
- Designed a **loan default prediction** pipeline using time-series feature engineering, building an early-warning system that flags at-risk customers 30 days in advance with 75% accuracy

Research Assistant

Jan 2024 - Dec 2024

Quantum Robotics & Automation Research Group, MIST

- Developed end-to-end **AutoML** platform (AKI) used by 100+ researchers – covers data upload, preprocessing, visualization, model training, and deployment with a no-code interface
- Developed an AI-based assistive interface achieving 92% accuracy in **eye-movement tracking**, enabling voice- and hand-free communication for paralyzed and post-surgery patients
- Led 3 AI research projects contributing to publications and mentored 7+ teams in ML/DL and computer vision

Education

BSc in Computer Science and Engineering

Feb 2019 - Dec 2023

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

- Thesis: Development of an intelligent system for eye motion tracking and action prediction

Technical Expertise

AI & LLM Systems	Agentic AI, RAG Pipelines, Multi-LLM Orchestration, Custom MCP Servers, Fine-Tuning (LoRA/QLoRA), Vector Search & Embeddings, Multi-Tenant Architecture
Frameworks & APIs	LangChain, LangGraph, LlamaIndex, FastAPI, REST & GraphQL, SSE/WebSocket Streaming, PyTorch, TensorFlow, Scikit-Learn, OpenCV
MLOps & Cloud	AWS (ECS, EC2, RDS, S3, Cognito, ElastiCache), Docker, CI/CD (GitHub Actions), Model Deployment, Self-Hosted LLMs (vLLM), Monitoring (LangSmith)
Databases	PostgreSQL (pgvector), Neo4j (Graph DB), Redis, MongoDB, Weaviate
Languages	Python, C/C++, SQL, Bash, Git

Key Projects

Business Intelligence Agent	<i>Demo</i>
Enterprise AI agent with full access to company databases and documents – investigates business questions, forecasts sales trends, generates data-driven strategic insights for executive leadership, and automates routine business operations	
Stack: LangChain · LangGraph · FastAPI · FastMCP · PostgreSQL · pgvector · Docker · LLMs · SQL	
Interactive AI Tutor	<i>Demo</i>
AI-powered learning platform where students learn through interactive simulations with an integrated AI agent – supports real-time visualization, simulation control, and conversational Q&A	
Stack: LangChain · FastAPI · Agentic AI · Real-Time Streaming	
Eye Motion Tracking Assistive System	<i>Demo</i>
Computer-vision system enabling paralyzed users to operate a computer hands-free with 92% gaze-tracking accuracy	
Stack: Scikit-Learn · TensorFlow · Computer Vision · Python	
Controlling Robotic Hand from Remote Distance	<i>Demo</i>
Gesture-controlled robotic hand using computer vision for remote teleoperation	
Child Companion AI Robot	<i>Demo</i>
IoT companion robot with vision-based child interaction and a Flutter parental monitoring app	

Awards & Recognition

Champion - AI Hackathon 2025 (Healthcare)	<i>Certificate</i>
Bangladesh Innovation Conclave — Led team of 3 to design an AI healthcare application	
Global Champion - University Rover Challenge	
The Mars Society — Competed against 20+ international teams worldwide	
1st Runner-Up - BEAR Summit 2025 (AI Demo)	<i>Certificate</i>
Bangladesh National Semiconductor Symposium — Submission code AI-047	
EDGE-AI Certification (Top 5%)	<i>Certificate</i>
World Bank-funded program with Bangladesh Computer Council — 160-hour enterprise AI specialization	